MIL-P-48154B (AR)
AMENDMENT 2
20 October 1986
SUPERSEDING
AMENDMENT 1
19 March 1985

MILITARY SPECIFICATION

PROPELLANT M1 FOR USE IN CARTRIDGE
105MM, TPDS-T, M724A1

This Amendment forms a part of Military Specification MIL-P-48154B (PA) dated 3 December 1976 and is approved for use within the US Army Armament Munitions and Chemical Command and is available for use by all Departments and Agencies of the Department of Defense.

PAGE 2

* 3.1, line 3: Delete "0.026" and substitute "0.023".

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3.2.2, line 2: Delete "10.0 fps (3.05 mps)" and substitute "12.0 fps".

PAGE 4

* 4.3.2.3: Delete in its entirety and substitute the following:

"4.3.2.3 Velocity Uniformity (70°F) (see 3.2.2). If the product of the factor (see 6.3) times the velocity standard deviation of the rounds tested in accordance with 4.4.1.2 exceeds the applicable requirement, apply referral criteria in accordance with 4.4.1.4. If the referral criteria cited do not apply, the lot shall be rejected."

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* 4.4.1.3: Delete in its entirety and substitute the following:

"4.4.1.3 Retest criteria. All original and retest data shall be submitted to Commander, Armament Research, Development and Engineering Center, US Army Armament, Munitions and Chemical Command, ATTN: AMSMC-QAR-R(D), Dover, NJ 07801. A lot shall be retested whenever any of the following conditions exist:

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a. The final adjustment in charge weight exceeds 2.0 oz of the charge weight used in uniformity firings retest for 70°F uniformity in accordance with 4.4.1.2.

b. If for any reason, the Proving Ground considers that test conditions have detrimentally affected test results, additional rounds as required shall be tested."

* 4.4.1.4: Delete in its entirety and substitute the following:

"4.4.1.4 Referral criteria. A test lot shall be referred to the procuring activity for disposition whenever any of the following conditions exist or whenever the proving ground deems it necessary:

a. Calibration rounds fired in any test phase fail to comply with 3.2.2 of this specification.

b. Calibration correction applied exceeds 25 fps.

c. Number of test or calibration results utilized in calculations is less than specified.

d. Any unusual occurrences during ballistic testing, such as excessive delays between rounds, poor projectile flights, etc. Each such event shall be reported in detail on the applicable firing record.

e. Test lot fails and any test value is an outlier. The following test shall be performed to determine if a single test result is an outlier. With a sample size of 10 at a significance level of 0.05, the critical value (T_c) is 0.477. Calculate a test value (T_t) as follows:

For Low Outlier

$$T_t = \frac{(X_2 - X_1)}{(X_9 - X_1)}$$

For High Outlier

$$T_t = \frac{(X_{10} - X_9)}{(X_{10} - X_2)}$$

X_1 or X_{10} is an outlier if T_t is greater than T_c.

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Where:

X_1 = Test result being tested as low outlier.
X_2 = Test result adjacent to X_1, after all the test results are ordered from smallest to largest.
X_9 = Test result adjacent to the largest test result.
X_{10} = Test result being tested as high outlier."

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* 6.8: Delete in its entirety.

PAGE 8

* 6.9: Delete in its entirety.

PAGE 9

* Figure 1: Delete in its entirety.

The margins of this amendment are marked with an asterisk or vertical lines to indicate where changes (additions, modifications, corrections, deletions) from the previous amendment were made. This was done as a convenience only, and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous amendment.

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